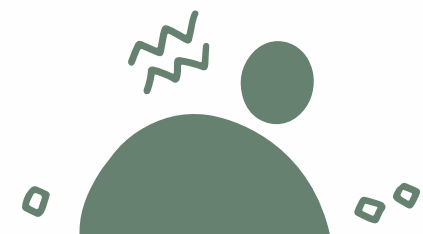
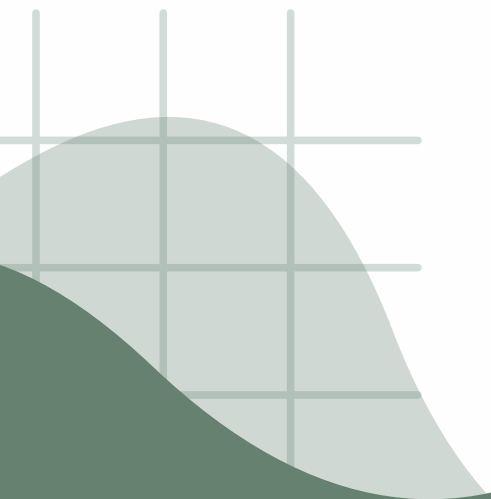




HTX TECHNICAL

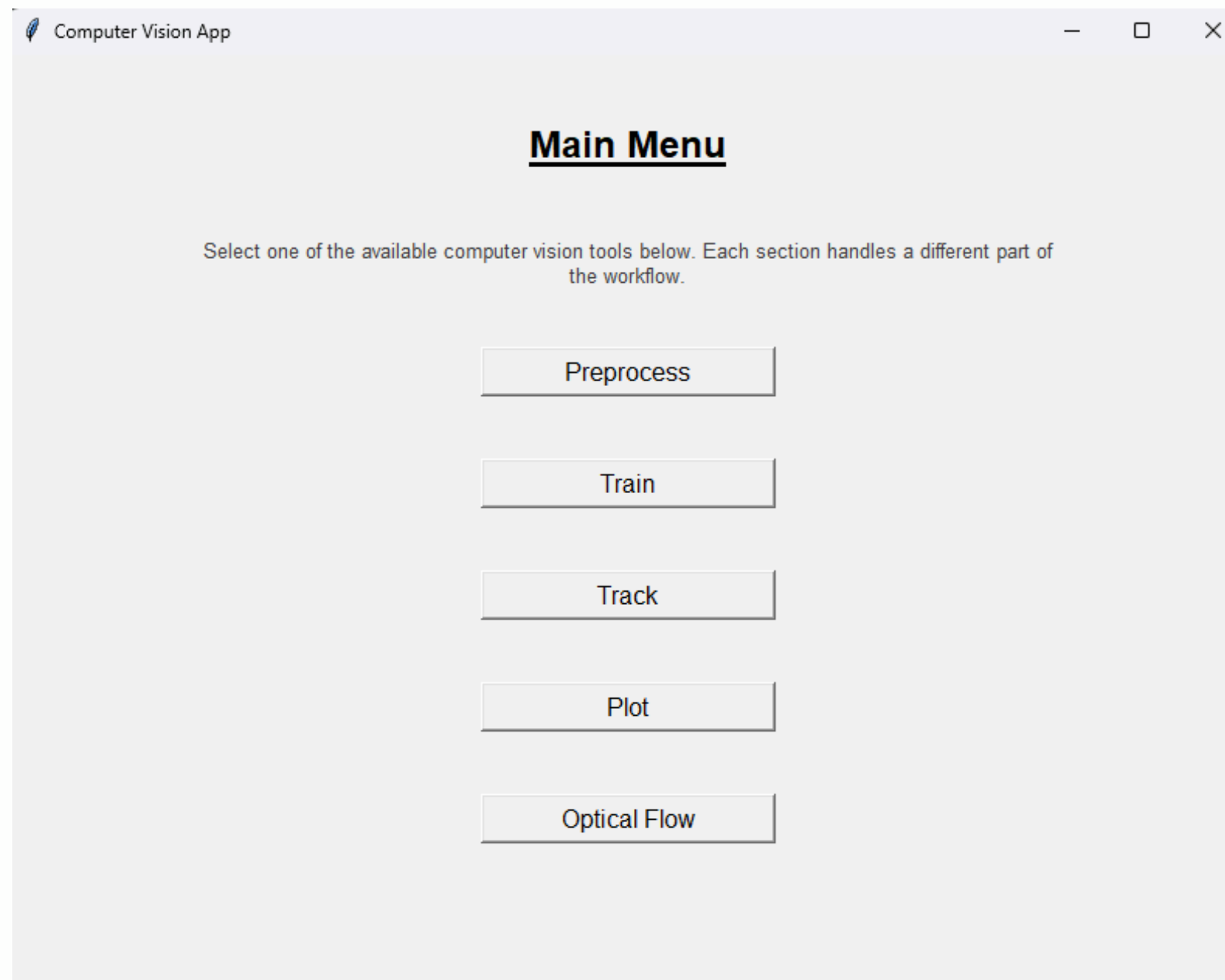
TEST

By Jun Kai





COMPUTER VISION

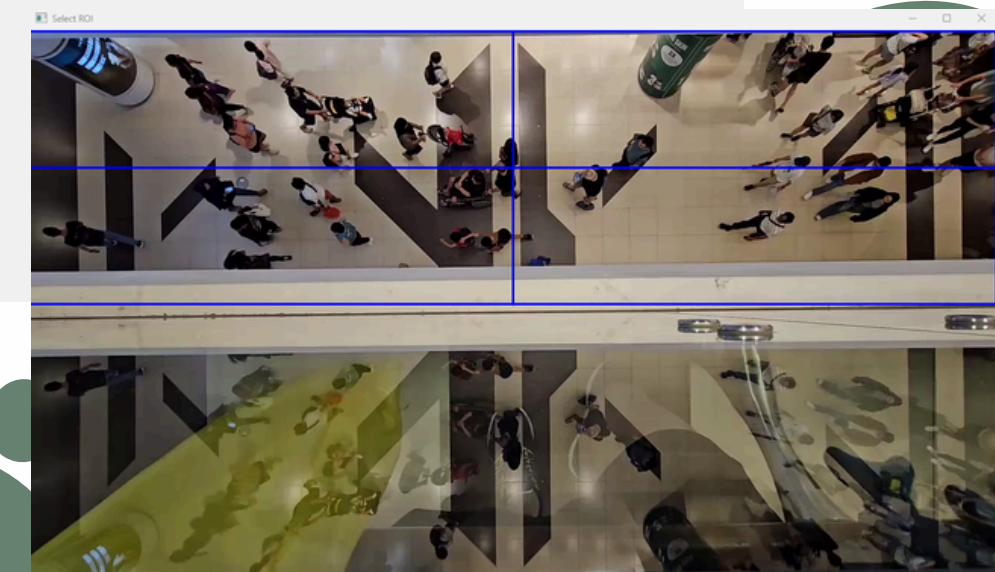
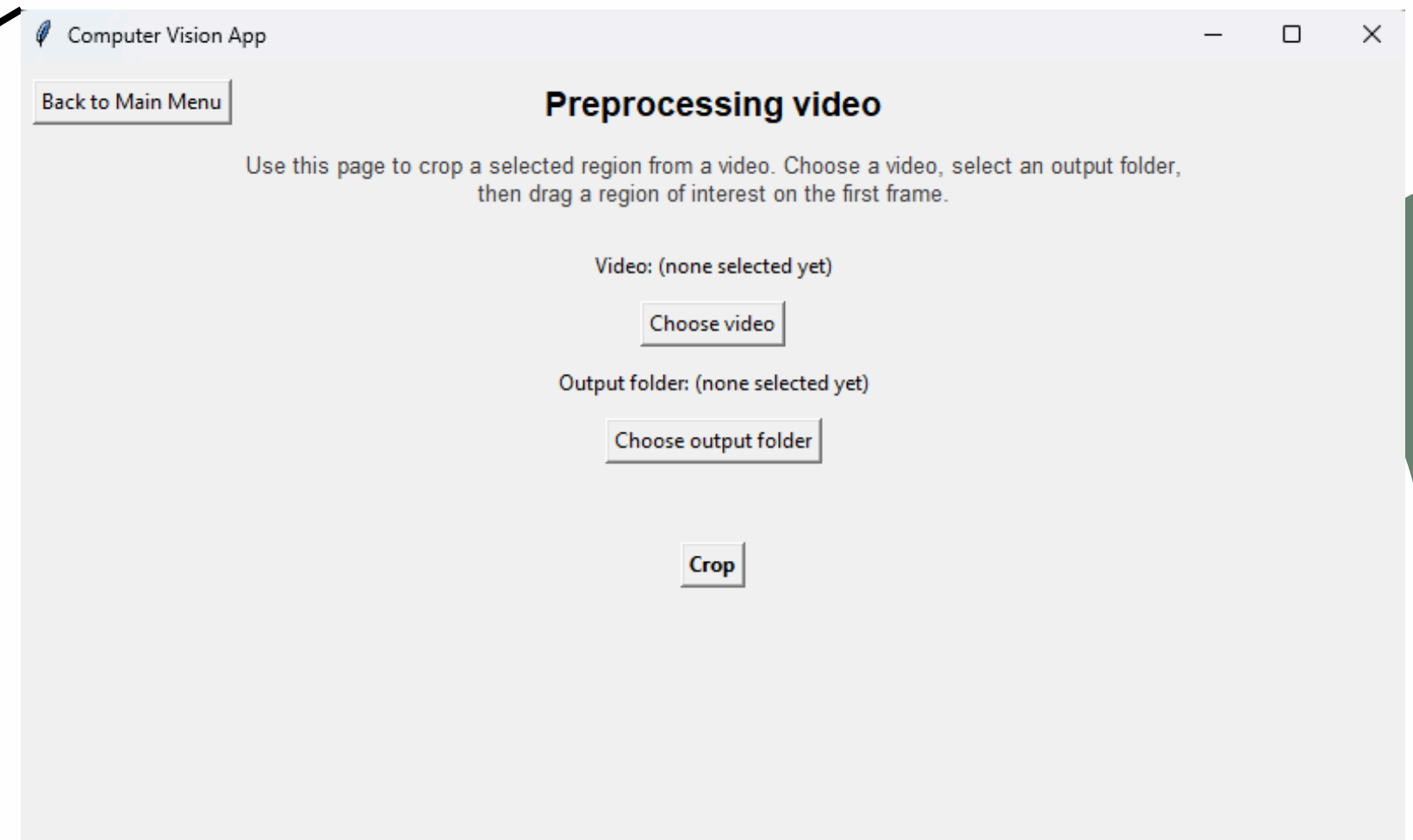
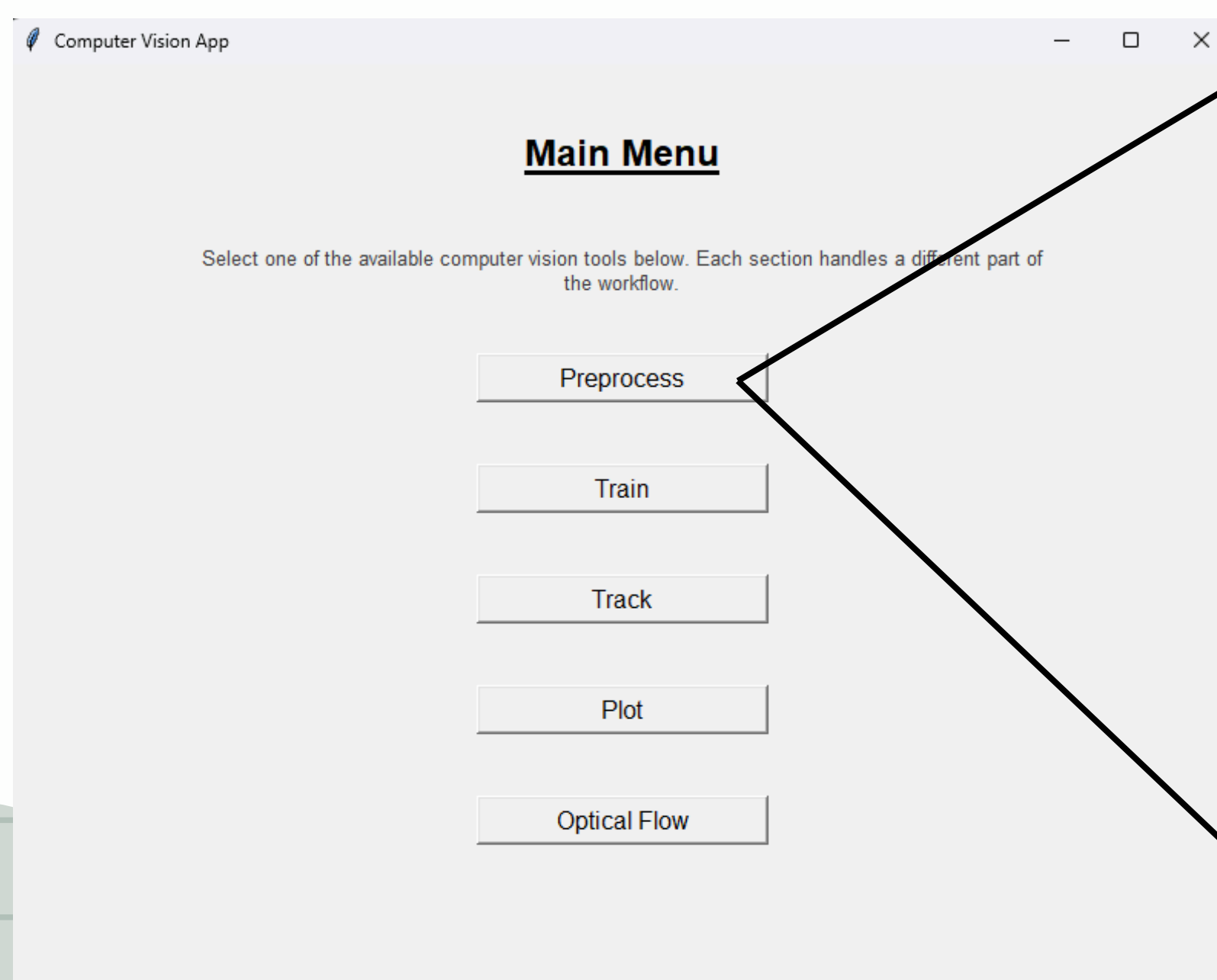


I created a mini UI to perform several computer vision task for the extraction of data to use for simulation



COMPUTER VISION

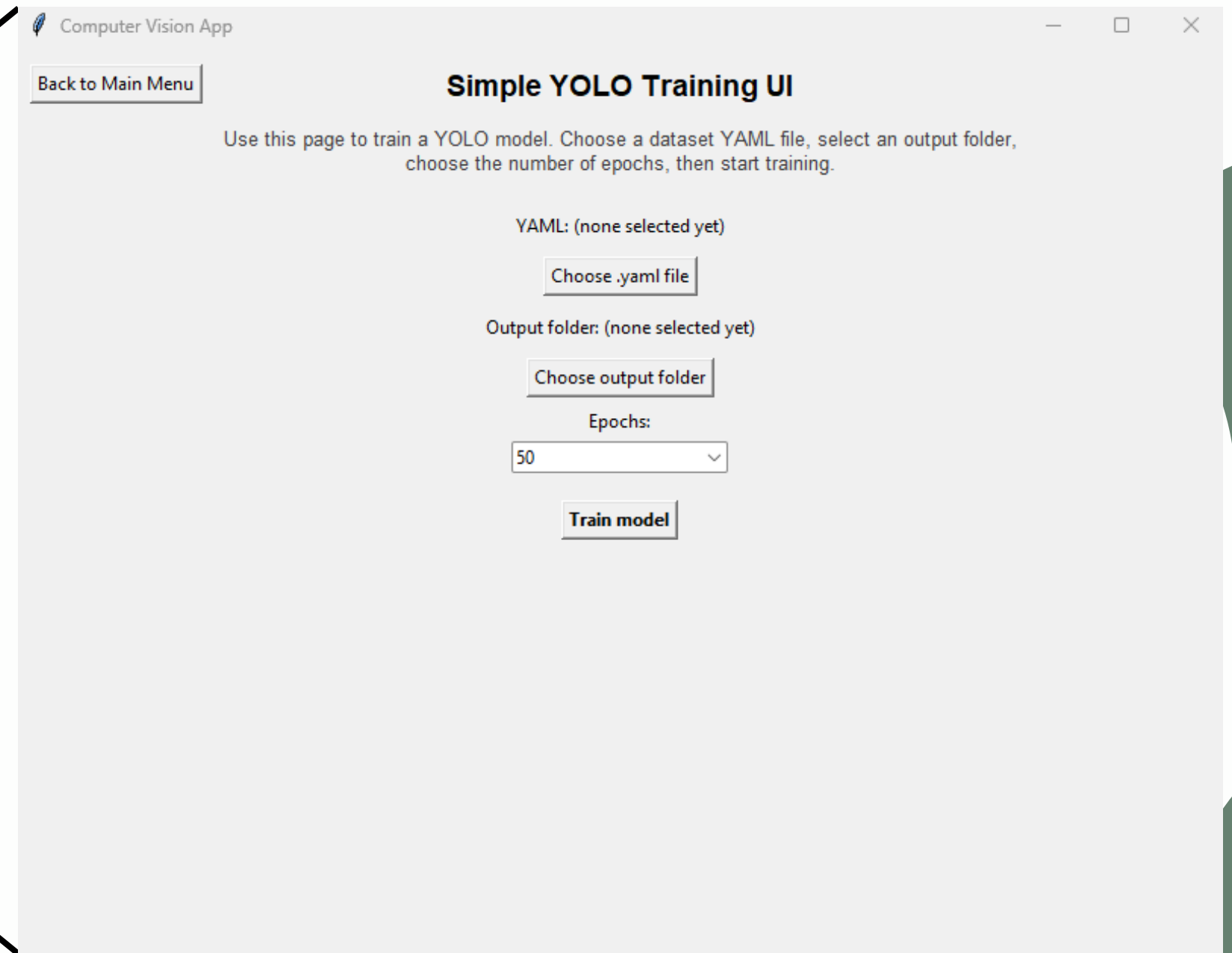
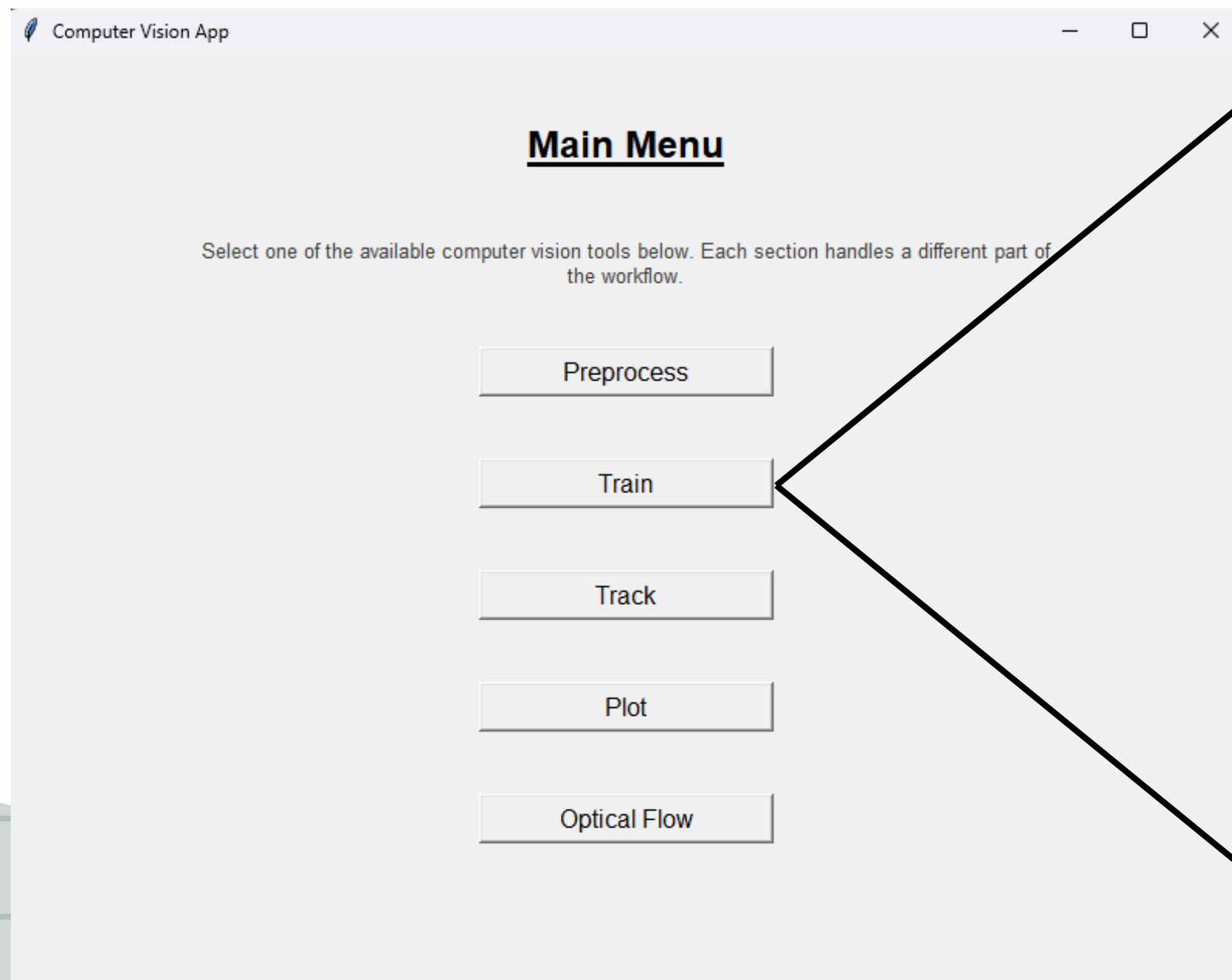
Cropping of videos by dragging a rectangle over chosen video to isolate the region of interest





COMPUTER VISION

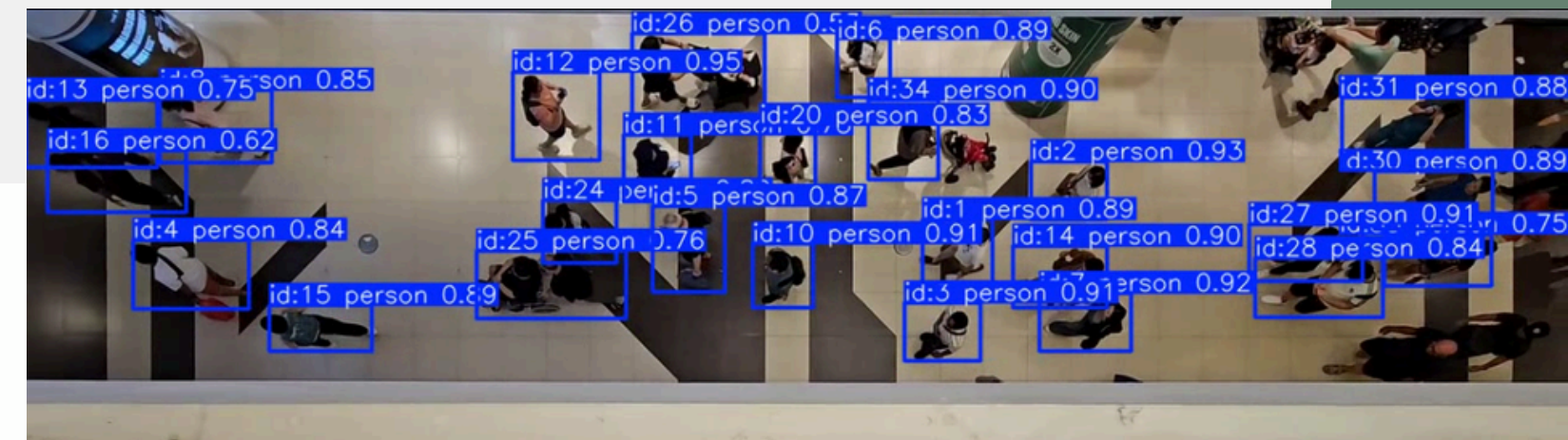
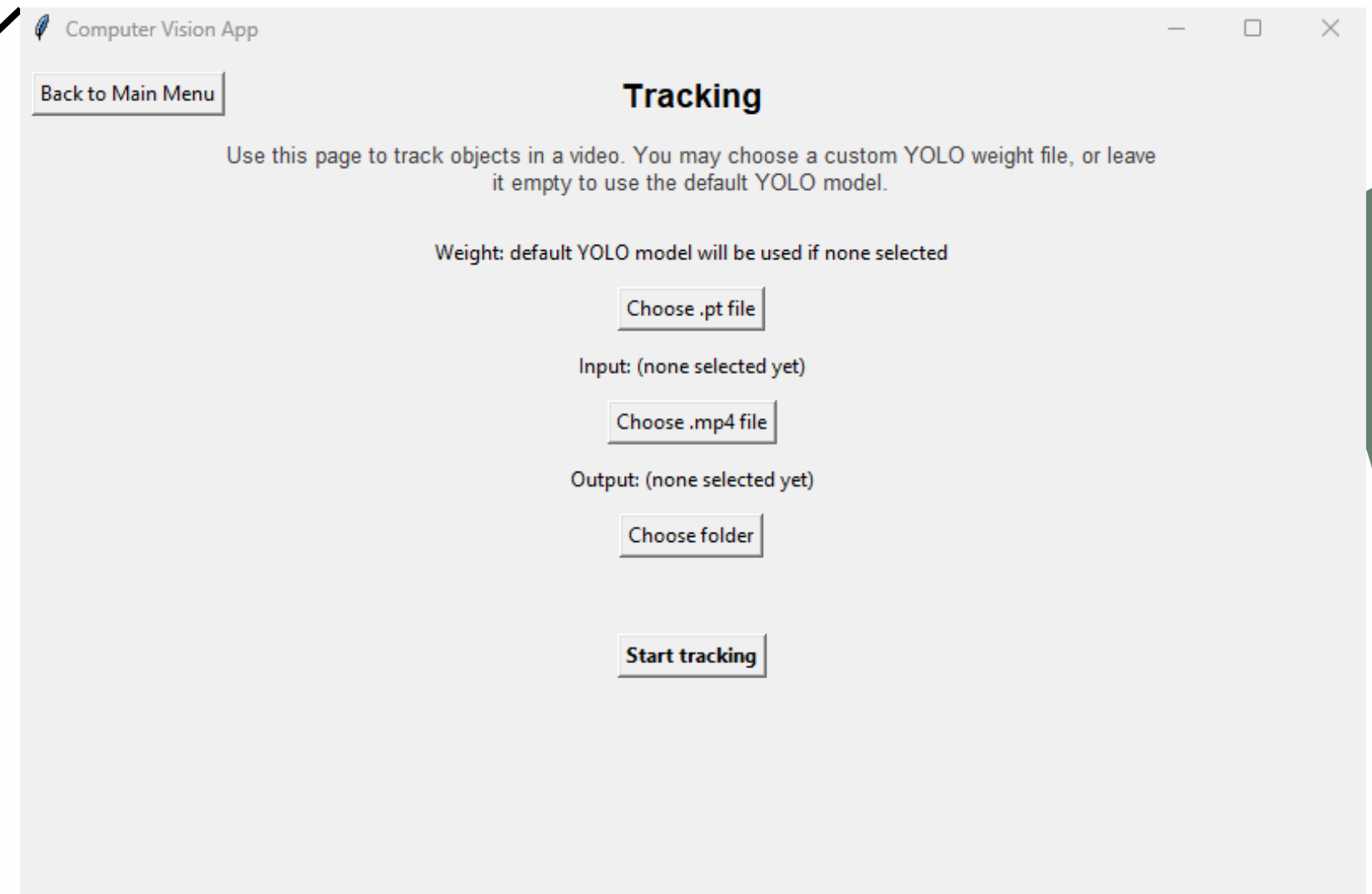
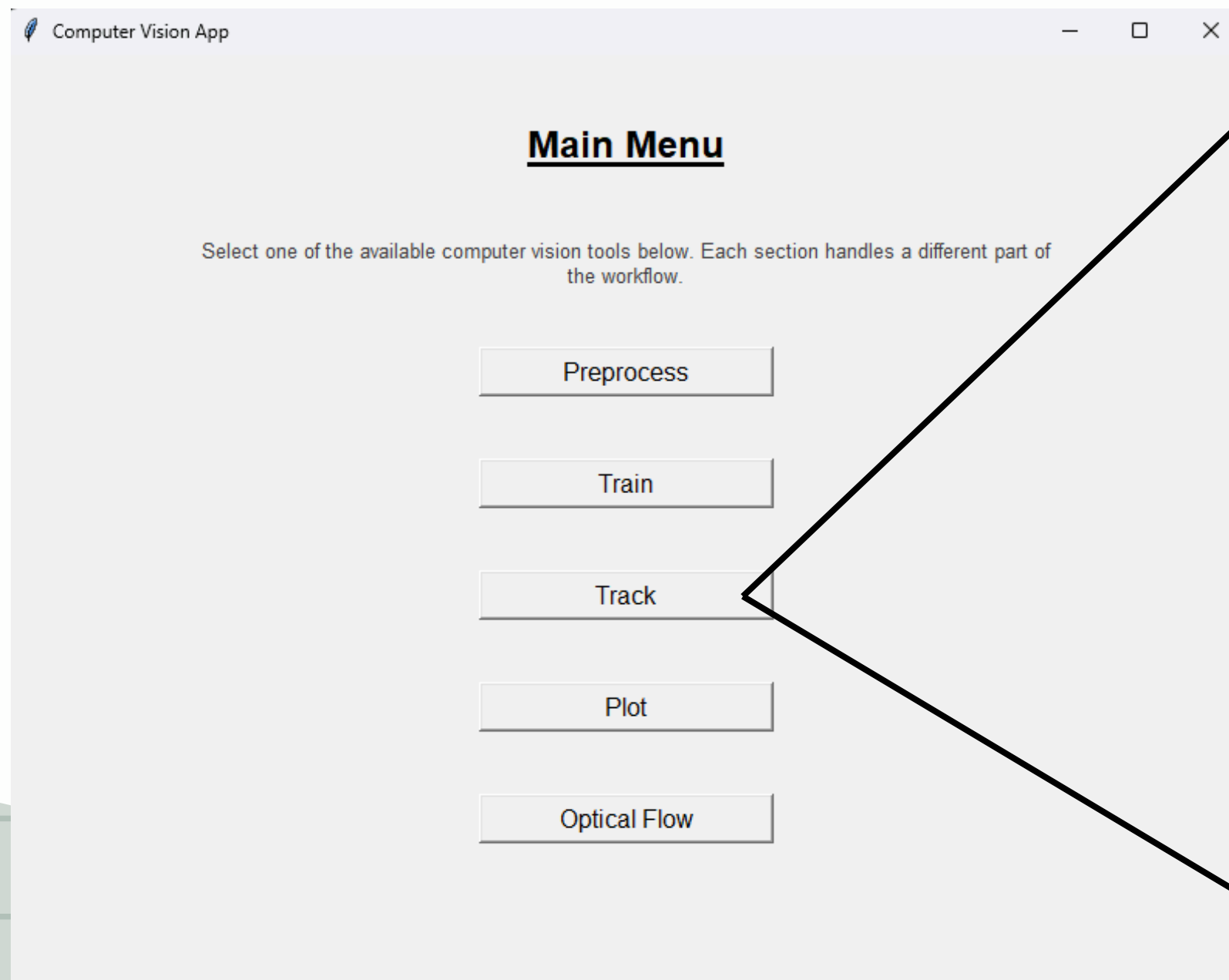
Training models given the annotated images and the .yaml file





COMPUTER VISION

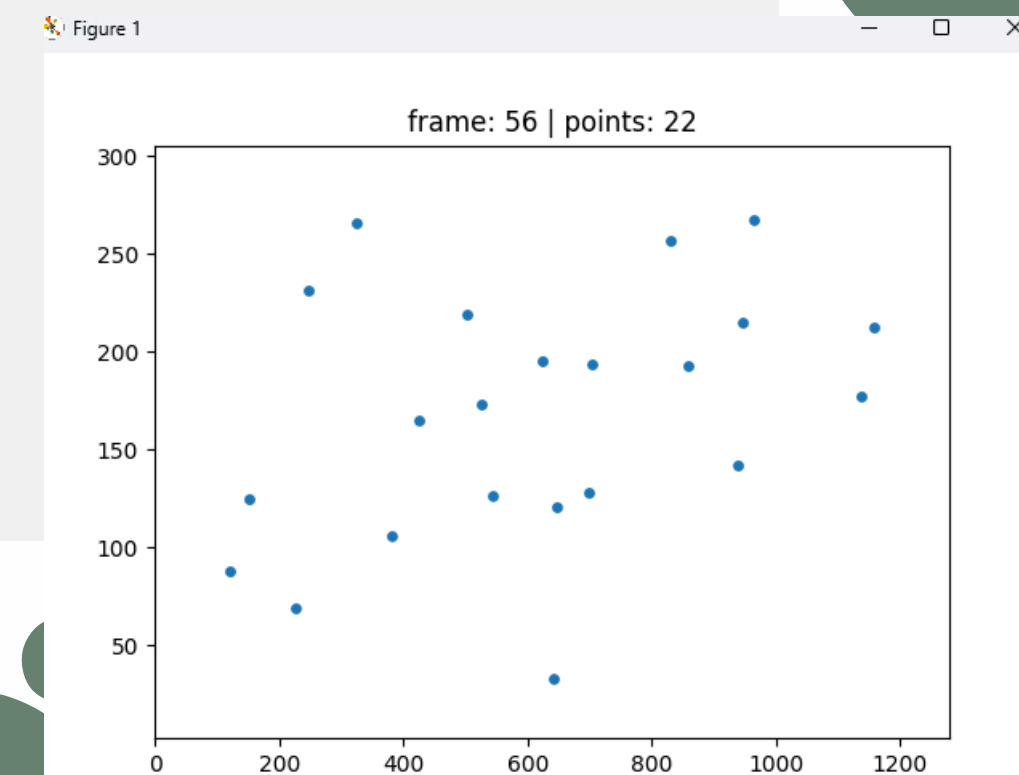
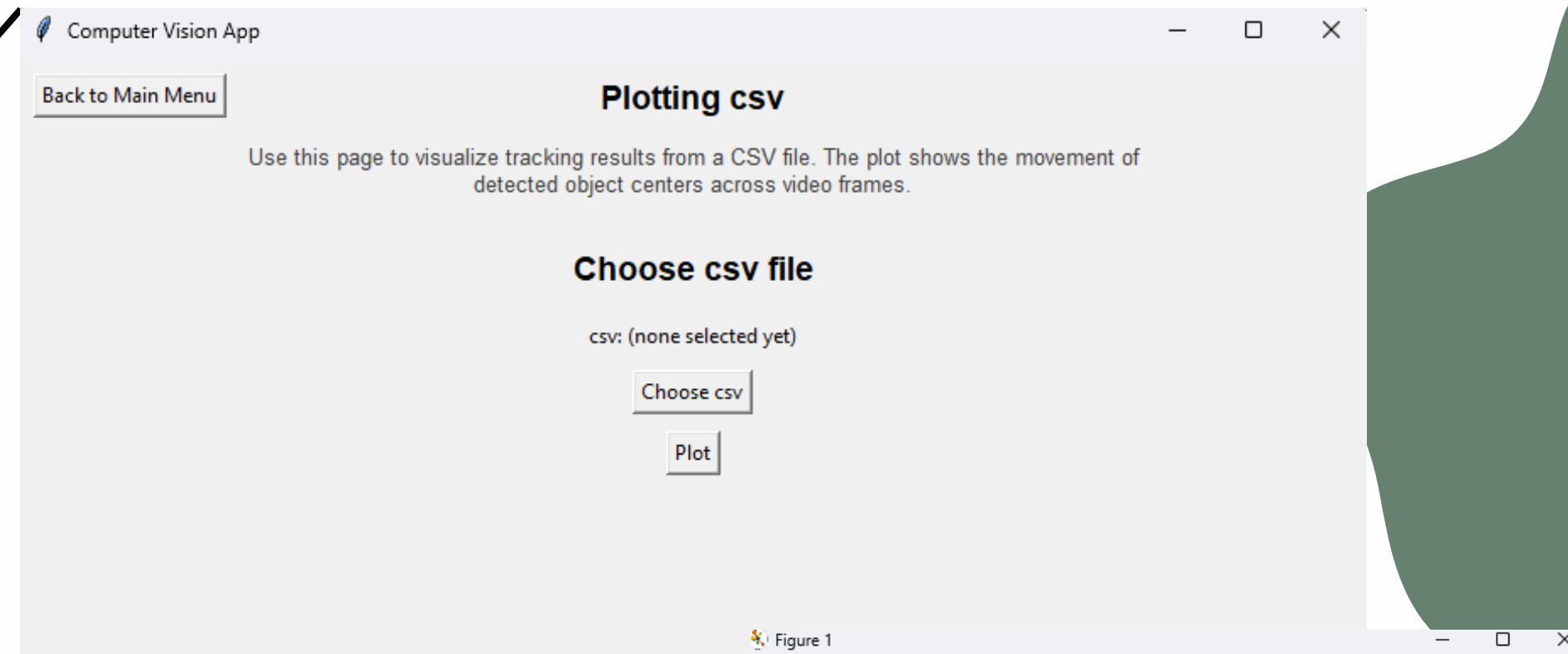
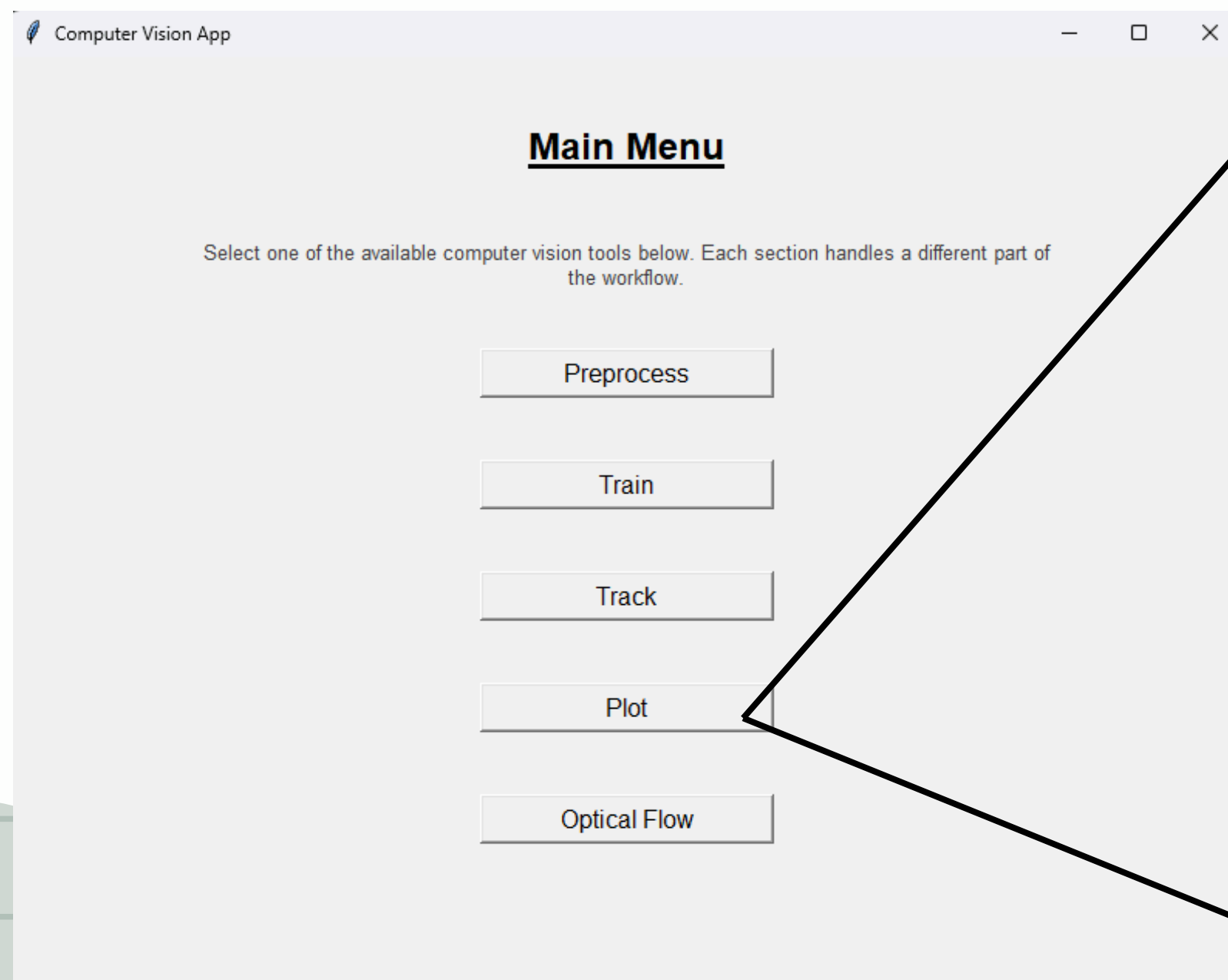
Tracks people in a video and outputs a csv file of the bounding box, centre of bounding box and speed of every ID per frame. It also gives the count of people in each frame





COMPUTER VISION

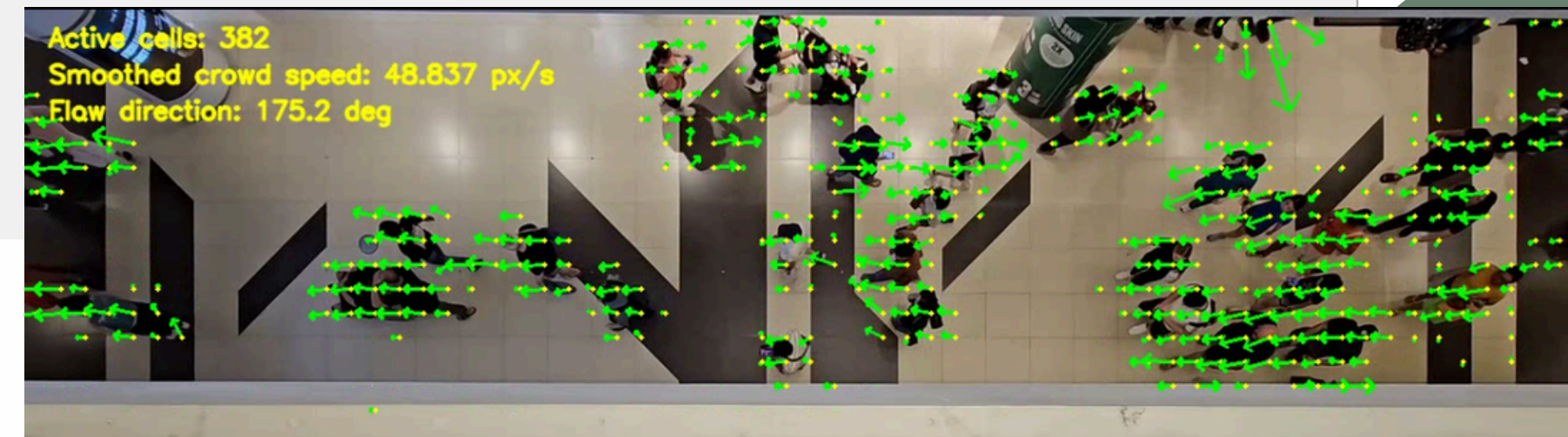
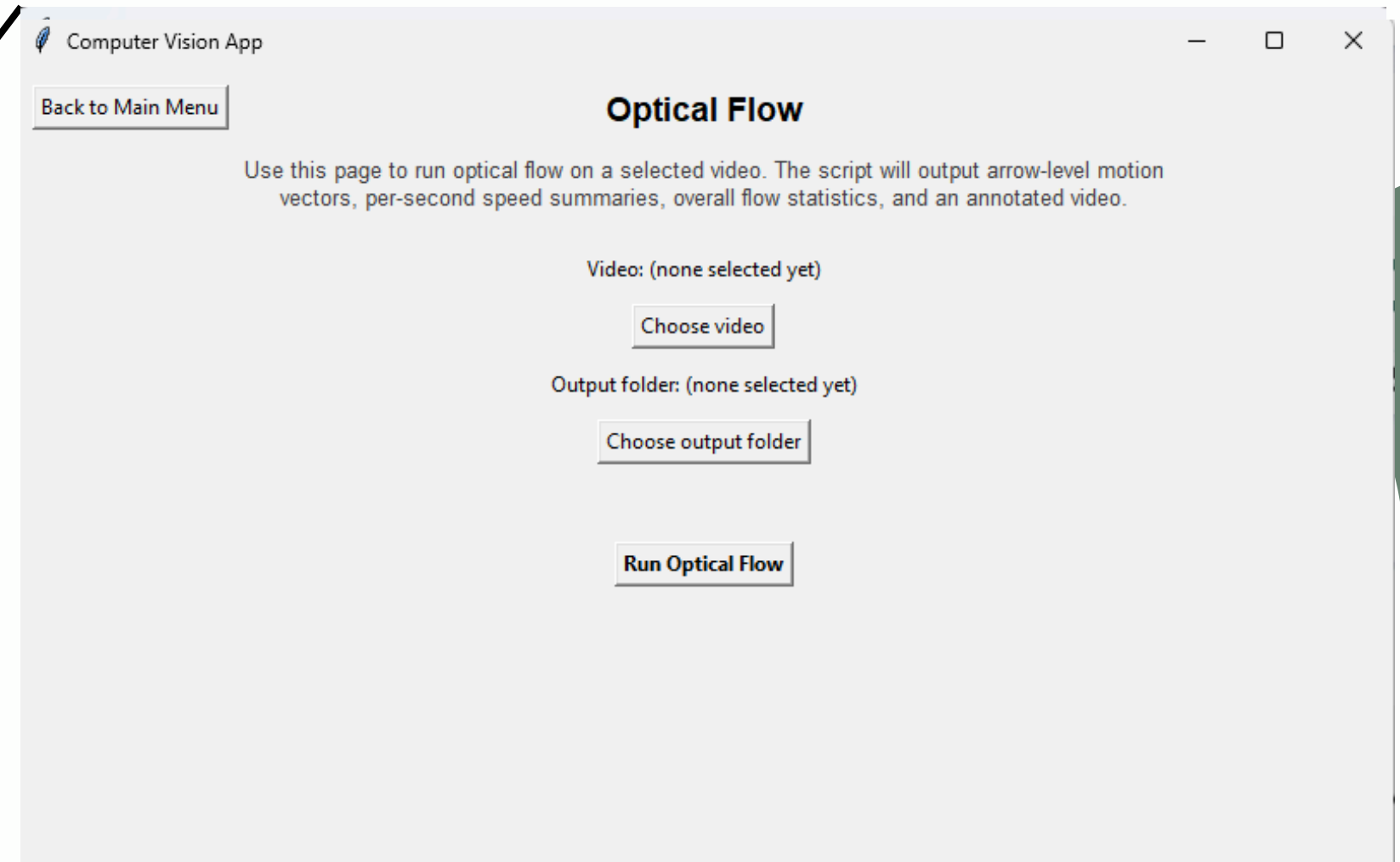
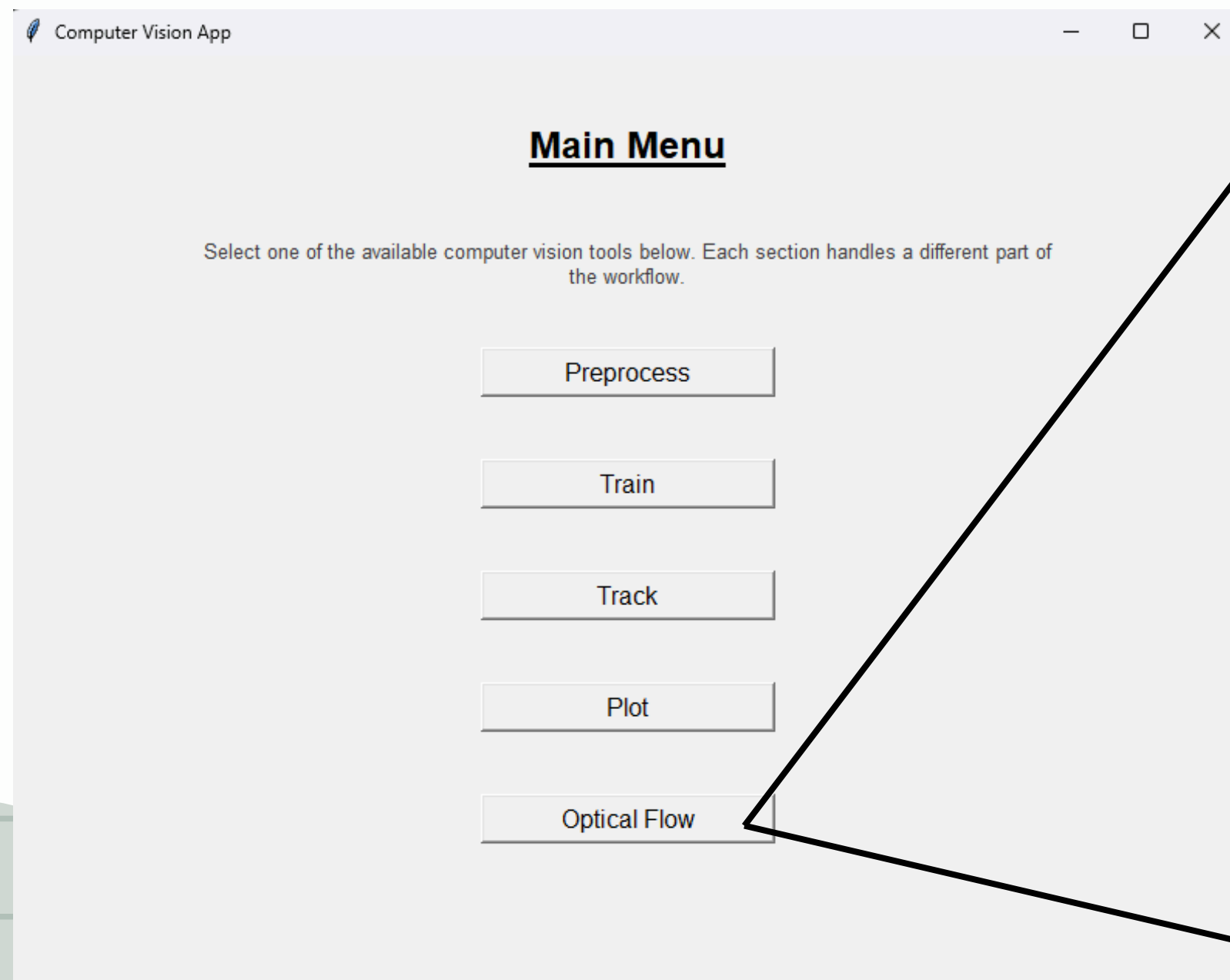
A simple 2D map of points of people given the respective csv files





COMPUTER VISION

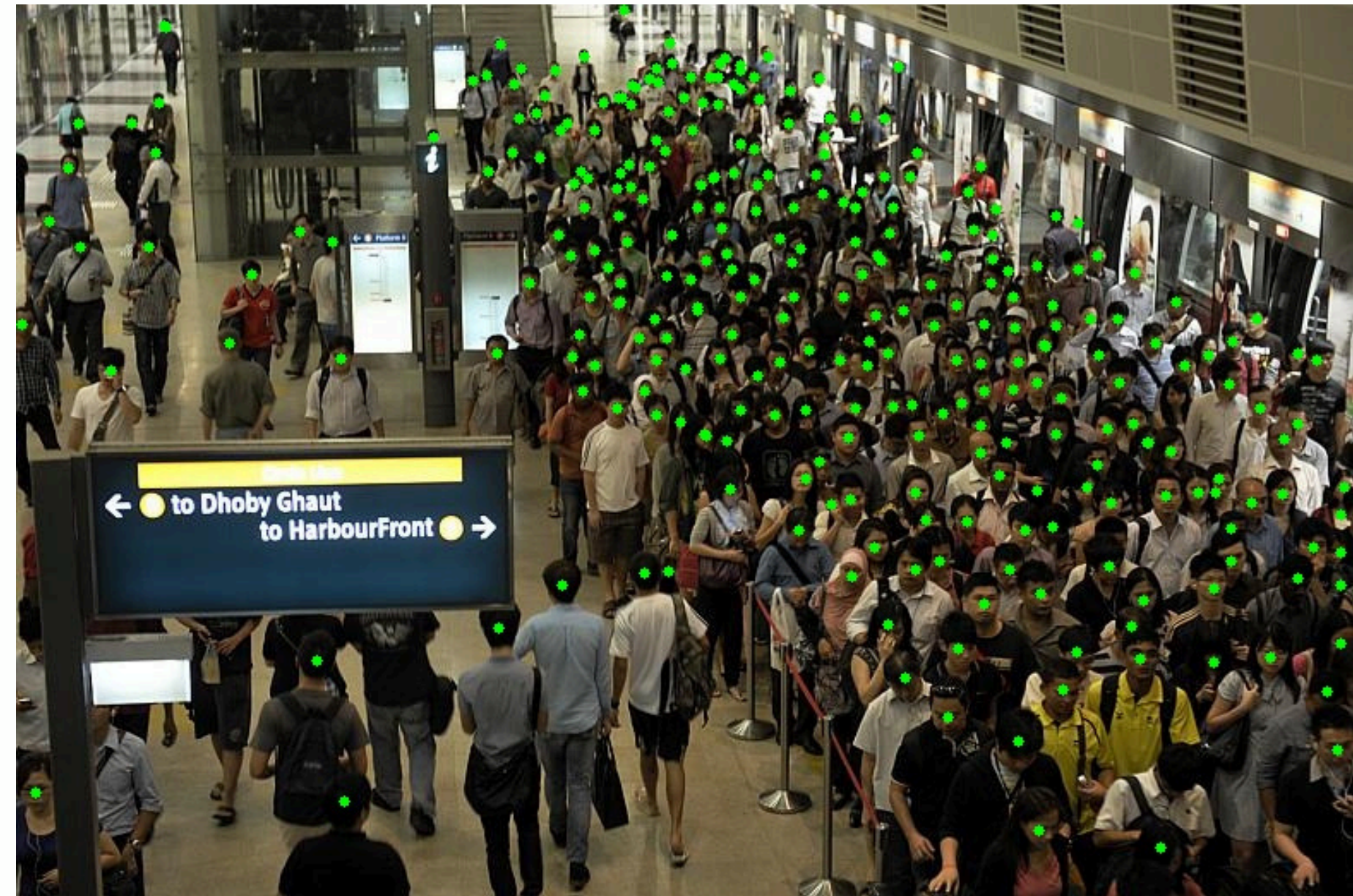
Optical flow to calculate speed of a denser crowd where tracking is computationally expensive





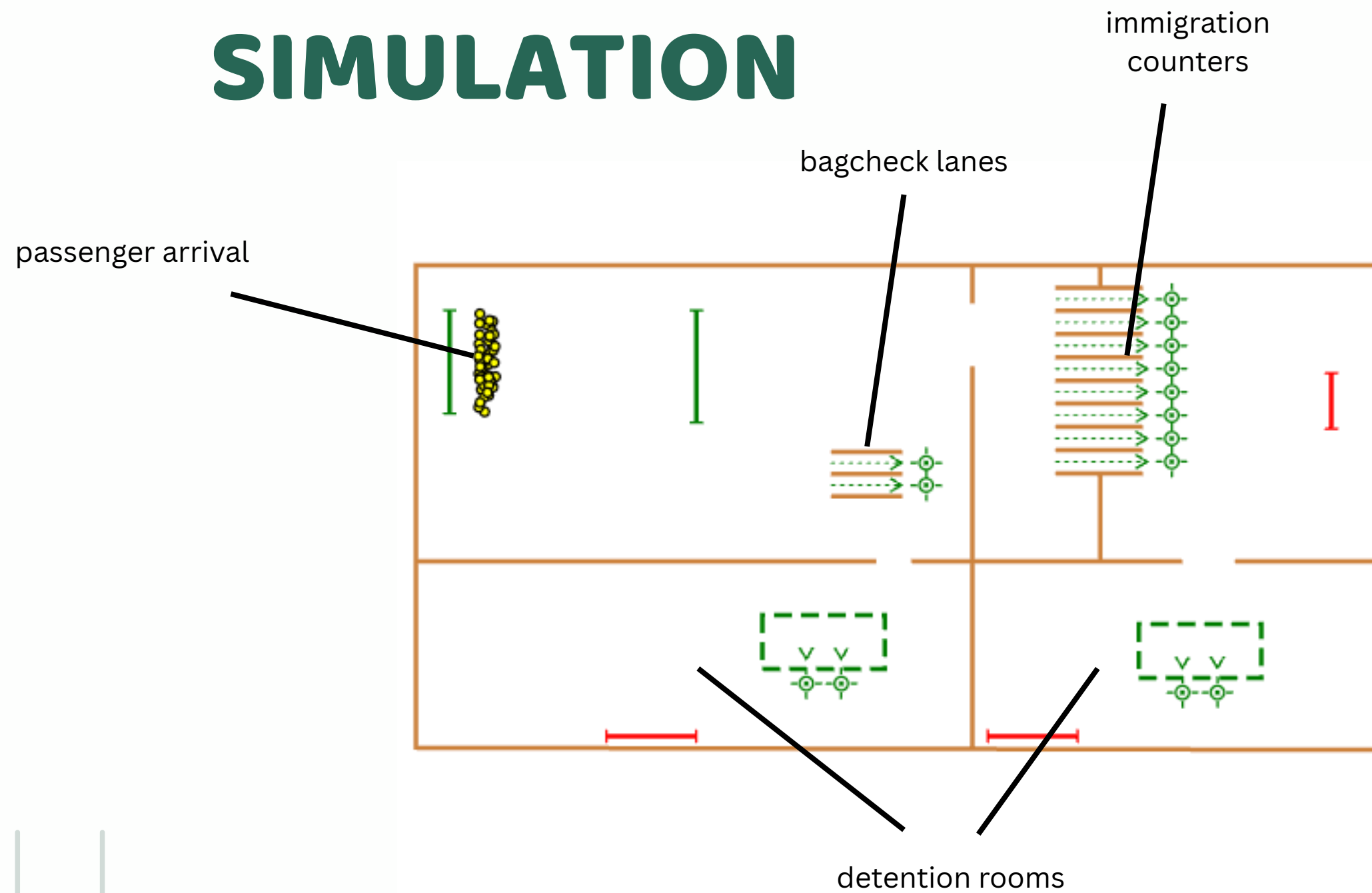
COMPUTER VISION

There is still a script that has not been added to the UI but is functioning already which is called Point Query Transformer. Useful for counting high density crowds.





SIMULATION



customisable dashboard

Completed travellers: 0

Security queue length: 0

Immigration queue length: 0

Security counters: 2 available / 2

Immigration counters: 8 available / 8

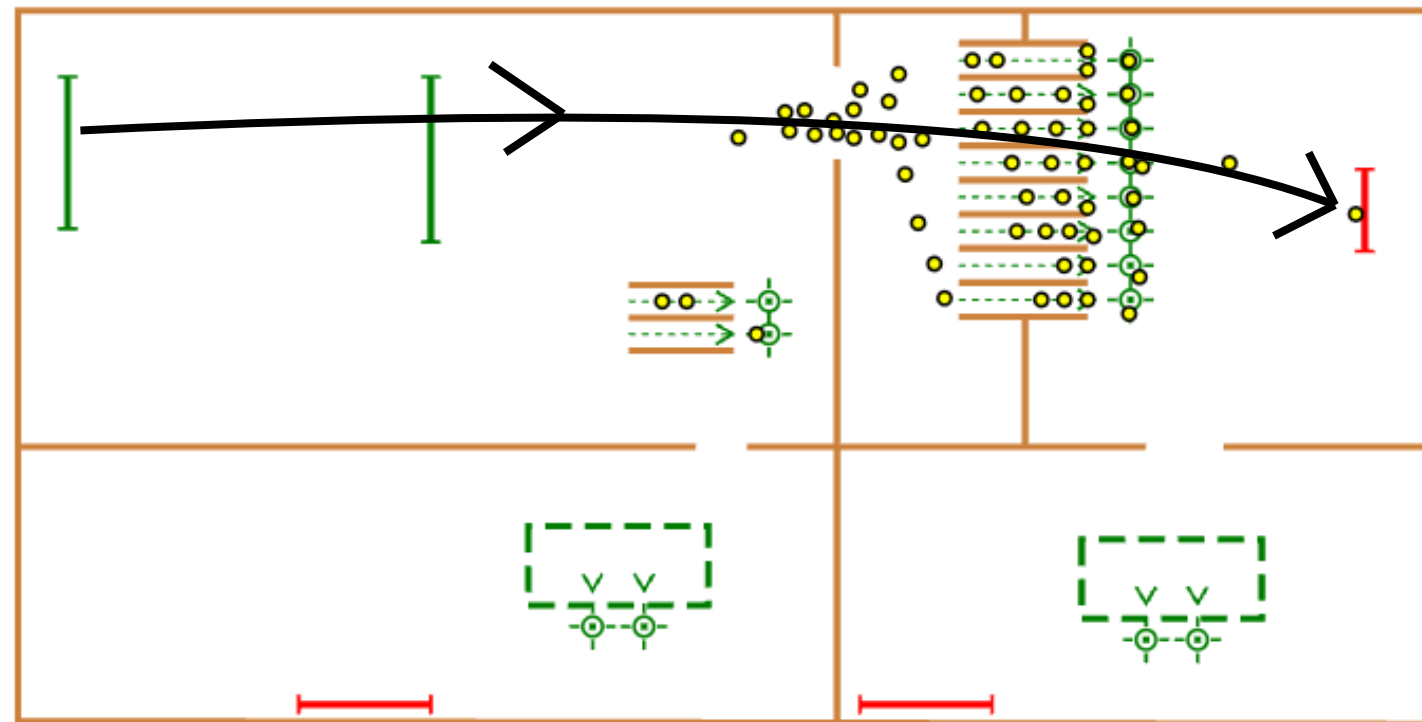
Avg security wait: 0.0 min

Avg immigration wait: 0.0 min

Avg total time: 0.0 min



SIMULATION

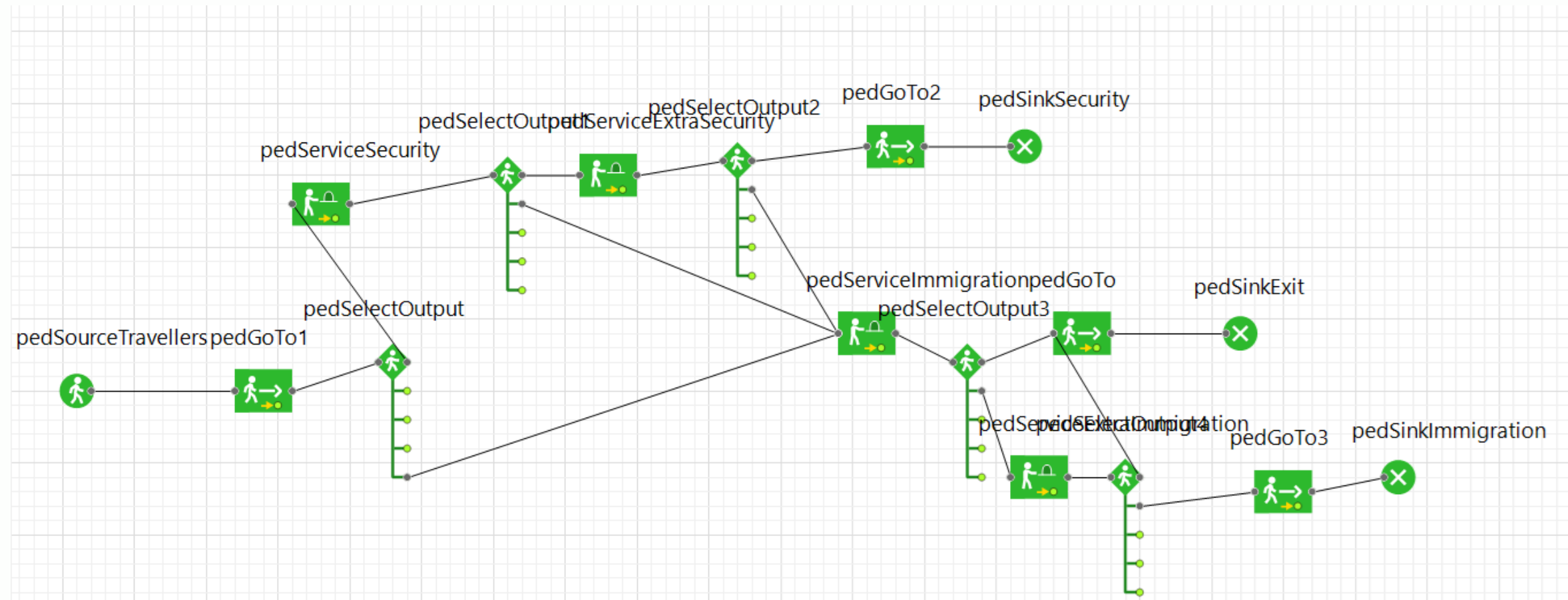


Completed travellers: 98
Security queue length: 2
Immigration queue length: 45
Security counters: 2 available / 2
Immigration counters: 1 available / 8
Avg security wait: 0.13 min
Avg immigration wait: 1.37 min
Avg total time: 1.98 min

The arrow line shows the most common path for the pedestrians to take unless singled out for bag check or if they have issue at the immigration counter



SIMULATION



This is the logic behind the pedestrians movement and most of the parameters are customisable such as their walking speeds, how long each service takes, the arrival timing of buses and even the frequency of bag checks to find the maximum number of checks at the least cost to overall waiting time

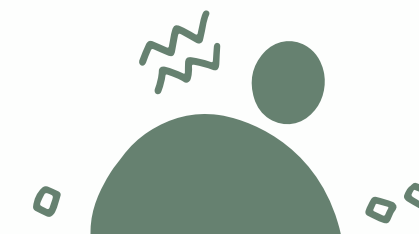


SIMULATION

⚡ eventVehicleArrival	🔗 maxBusPassengers
	🔗 minBusPassengers
	🔗 busIntervalMinutes
	🔗 numSecurityCounters
	🔗 numImmigrationCounters
	🔗 securityCheckProbability
	🔗 extraSecurityCheckProbability
	🔗 extraSecurityCheckFailProbability
	🔗 extraImmigrationCheckProbability
	🔗 extraImmigrationCheckFailProbability

Examples of some of the parameters that can be changed. functions can be added inside the simulation also for my customisable outputs/dashboarding

The data of each run is then saved onto a csv file to be analysed and compared between different parameters

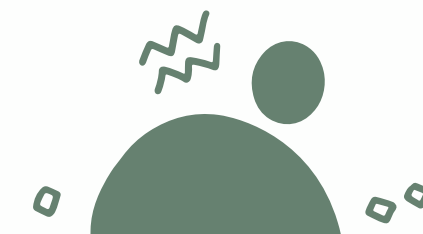
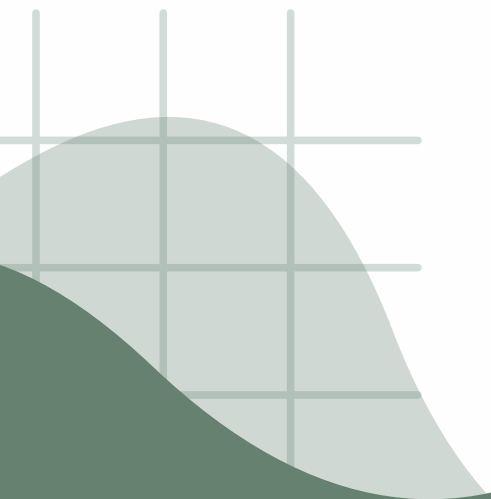




FUTURE IMPROVEMENTS

Find a way to link both of the task together where the data is fed directly into the simulation engine

Using the simulation come up with some constructive insights as to how the simulation can help with maximising resource efficiency





THANK YOU

